

- 1 (a) A truck is accelerated from rest by a constant driving force. It experiences a drag force which is proportional to its speed. Eventually, the truck travels at a constant velocity.

Explain why the truck reaches a constant velocity.

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..... [2]

- (b) On Fig. 1.1 sketch a graph to show how the acceleration of the truck varies with time from the time when it was at rest to the time when it reaches the constant velocity.

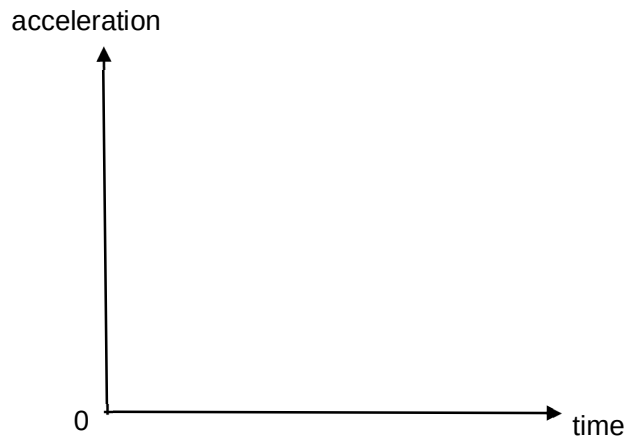
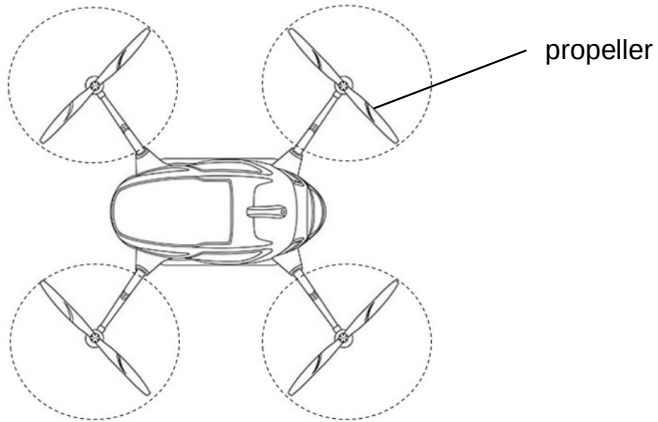


Fig. 1.1

[1]

- (c) Fig. 1.2 shows a drone that is able to hover by rotating the four horizontal propellers attached to its body. The propellers push the air downwards and impart the air with a certain speed. The mass of the drone is 0.50 kg and each of the propellers has a radius of 15.0 cm.



Top view of a drone

Fig. 1.2

- (i) Calculate the upward force by the air on the drone.

upward force = N [1]

- (ii) State, with a reason, the magnitude of the downward force exerted by all of the drone's propellers on the air.

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..... [2]

(iii) The density of air is 1.3 kg m^{-3} .

Determine using Newton's second law of motion the speed imparted to the air by the propellers.

speed = m s^{-1} [4]